VOL. I · NO. 10 MONDAYS & THURSDAYS

THE LAUNCH REPORT

Rocket launches, reentries and the business of spaceflight, worldwide

THURSDAY, NOVEMBER 12, 2026 ALL TIMES EASTERN STANDARD TIME (EST)

THE WEEK IN LAUNCH

Vulcan returns and Starship goes for a double catch on Friday

Two orbital launches flew worldwide since Monday's issue, and both reached orbit: a Long March 2D from Jiuquan with the Chinese Academy of Sciences' Pathfinder-1 science satellite on Wednesday morning, and a Falcon 9 from Vandenberg with the classified USSF-627 mission late Wednesday night, two days later than planned. That brings 2026 to 262 orbital launch attempts and 255 successes, counting GCAT through Sunday and the two launches since.

Friday is the big day. United Launch Alliance's (ULA) Vulcan flies for the first time in more than nine months with 40 Amazon Leo satellites at 12:54 p.m., and the FAA's airspace plan now carries Starship Flight 15 at 5:45 p.m., with a first attempt to catch both the booster and the ship (FAA). Rocket Lab's 100th Electron follows early Saturday. South Korea's military rocket demonstration is off again after a diver died working on its launch barge, and NASA's twin ESCAPADE probes are now headed for Mars.

I · RECENT LAUNCHES & RETURNS

A science pathfinder for China; a second classified Falcon in six days

One Chinese and one American launch, both successful, between Monday morning and Wednesday night.

Long March 2D: CAS Pathfinder-1

A Long March 2D lifted off at 6:41 a.m. EST Wednesday (11:41 UTC) from Launch Area 94 at Jiuquan with Pathfinder-1, a Chinese Academy of Sciences (CAS) satellite bound for Sun-synchronous orbit (Launch Library 2). It tests key technologies for the academy's "Origins" space science program of four astronomy missions due by 2030, and carries a gamma-ray telescope pathfinder, a magnetometer, an upper-atmosphere lidar and a space weather monitor. The satellite "carries three subsatellites which will be deployed later," Jonathan McDowell noted; Launch Library 2 says they will fly in formation to demonstrate a low-frequency radio astronomy interferometer.

Falcon 9: USSF-627

"Liftoff!" SpaceX posted as a Falcon 9 left Space Launch Complex 4E at Vandenberg at 11:36 p.m. EST Wednesday (04:36 UTC Thursday) on USSF-627, a classified U.S. Space Force mission listed in Monday's FAA plan as "R-5." First stage B1103, on its seventh flight, landed on the drone ship Of Course I Still Love You (SpaceX). It was the second classified Space Force launch from Vandenberg in six days, after USSF-432 on Friday, and the 120th Falcon 9 launch of 2026, 85 of them for Starlink, Space.com reported. SpaceX has not described the payload; Launch Library 2 notes the rocket's drop zones match earlier Starlink Group 15 flights.

Sunday's SatNet batch catalogued

The nine SatNet satellites from Sunday's Long March 12 now appear in CelesTrak's catalog as Hulianwang Digui 232 to 240, at about 912 by 930 kilometers, inclined 50 degrees; the rocket's upper stage sits in a 791 by 911 kilometer orbit.

ON THE WIRE

Two from Vandenberg in six days. USSF-627 followed the classified USSF-432 mission from the same California base, which flew on Friday (Space.com).

ON THE WIRE

Next from Jiuquan. A Kuaizhou 11 from ExPace is listed for 2:50 p.m. EST Thursday from Launch Area 95A, with its payload not announced (Launch Library 2).

Three launches in 12 hours, starting Friday at lunchtime

Every orbital launch with a date between now and November 30. Times are EST with Coordinated Universal Time (UTC) below; NET means no earlier than.

DATE / TIME	VEHICLE	PROVIDER	MISSION / PAYLOAD	SITE	ORBIT	STATUS	SOURCES
Thu Nov 12, 2:50 p.m. 19:50 UTC	Kuaizhou 11	ExPace	Payload not announced	Launch Area 95A, Jiuquan, China	—	TENTATIVE	Launch Library 2; NASASpaceflight
Fri Nov 13, 12:54 p.m. 17:54 UTC	Vulcan VC6L	United Launch Alliance	Vulcan Leo 1 (LV-01; FAA: "KV-01"): 40 Amazon Leo broadband satellites	SLC-41, Cape Canaveral, Fla.	LEO	CONFIRMED	ULA; FAA
Fri Nov 13, 5:45 p.m. 22:45 UTC	Starship	SpaceX	Flight 15: Booster 22 and Ship 42 with 26 Starlink V3 satellites; first attempt to catch both stages	Orbital Launch Pad 2, Starbase, Texas	LEO	CONFIRMED	FAA; Launch Library 2
Sat Nov 14, 12:30 a.m. 05:30 UTC	Electron	Rocket Lab	"The Mountain God Rises": radar satellite for iQPS; Electron's 100th mission	Launch Complex 1, Mahia, New Zealand	LEO	CONFIRMED	Rocket Lab; Launch Library 2
Sat Nov 14, 10:55 p.m. 03:55 UTC Nov 15	Falcon 9	SpaceX	USSF-317, a classified U.S. Space Force mission	SLC-4E, Vandenberg, Calif.	LEO	TENTATIVE	Launch Library 2
Wed Nov 18, 11:49 a.m. 16:49 UTC	Falcon 9	SpaceX	USSF-266, a classified U.S. Space Force mission	SLC-4E, Vandenberg, Calif.	Polar	TENTATIVE	Launch Library 2
Fri Nov 20, 6:00 a.m. 11:00 UTC	Long March 8A	CASC	SatNet low-Earth-orbit communications satellites (group not confirmed)	Commercial LC-1, Wenchang, China	LEO	TENTATIVE	Launch Library 2
NET Sun Nov 22	Ariane 64	Arianespace	VA271 (LE-04): 36 Amazon Leo broadband satellites	ELA-4, Kourou, French Guiana	LEO	PRELIMINARY	Arianespace; Launch Library 2
Tue Nov 24, 12:03 a.m. 05:03 UTC	Soyuz-2.1b	Roscosmos	Progress MS-36 cargo ship to the International Space Station	Site 31/6, Baikonur, Kazakhstan	LEO	TENTATIVE	Launch Library 2
Tue Nov 24, 2:00 a.m. 07:00 UTC	Long March 12	CASC	SatNet low-Earth-orbit communications satellites (group not confirmed)	Commercial LC-2, Wenchang, China	LEO	TENTATIVE	Launch Library 2
NET Tue Nov 24	Long March 2F	CASC	Shenzhou 24 crew to the Tiangong station	Launch Area 91, Jiuquan, China	LEO	PRELIMINARY	Launch Library 2
Thu Nov 26, 3:00 p.m. 20:00 UTC	Spectrum	Isar Aerospace	Third flight, carrying several satellites including R-Space's AT-Astra	Andøya Spaceport, Norway	SSO	TENTATIVE	European Spaceflight; Launch Library 2
Sun Nov 29, 10:49 a.m. 15:49 UTC	Falcon 9	SpaceX	Northrop Grumman's Cygnus NG-25 cargo ship to the space station	SLC-40, Cape Canaveral, Fla.	LEO	CONFIRMED	Launch Library 2

CONFIRMED date and time announced by the launch provider, its customer or the FAA. **TENTATIVE** specific date in published schedules or airspace and maritime notices, not yet confirmed by the operator. **PRELIMINARY** target range, or no-earlier-than date or month only.

Abbreviations: SLC, Space Launch Complex; SLC-4E, Space Launch Complex 4 East; ELA-4, Ariane Launch Area 4; LEO, low Earth orbit; SSO, Sun-synchronous orbit; CASC, China Aerospace Science and Technology Corp.; iQPS, Institute for Q-shu Pioneers of Space, a Japanese radar imaging company; USSF, U.S. Space Force.

Date changes

USSF-627. Monday, November 9, at 10:03 p.m.; flew Wednesday, November 11, at 11:36 p.m. (Launch Library 2). No reason given.

USSF-317. Thursday, November 12, at 11:23 p.m.; then Friday, November 13, at 11:09 p.m.; now Saturday, November 14, at 10:55 p.m. (Launch Library 2). No reason given.

USSF-266. Tuesday, November 17, at 11:58 a.m.; now Wednesday, November 18, at 11:49 a.m. (Launch Library 2). No reason given.

Mir demonstration. Wednesday, November 11, at 12:00 a.m.; then NET Friday, November 13; now no date. A diver in his 50s died Tuesday after underwater work on the barge set up to launch the rocket off Jeju, Korea JoongAng Daily reported, and the fourth test launch has been postponed indefinitely (SBS).

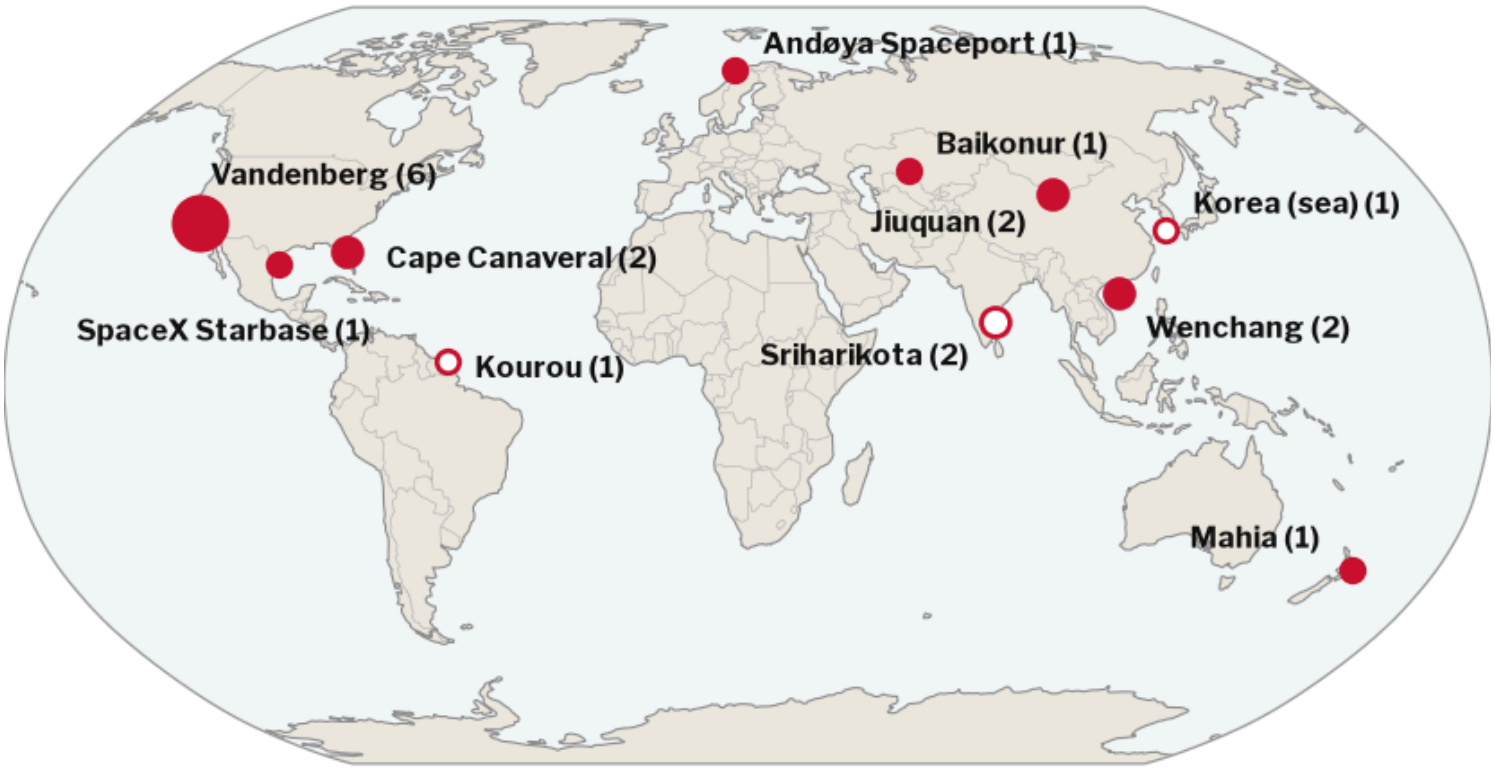
Starship Flight 15. Same 5:45 p.m. Friday target; the FAA plan now holds 5:45 to 7:40 p.m. on Friday and the same hours Saturday as backup (FAA).

Also listed for November without a specific date in Launch Library 2: Starlink 15-21, NROL-195 and OneWeb 21 on Falcon 9s from Vandenberg; and from India, the first uncrewed Gaganyaan test flight on a Launch Vehicle Mark 3 and a PSLV-XL with the TDS-01 technology demonstration satellite.

Further out

Europe's next Ariane 6 after VA271 is VA272 with the weather satellite Metop-SG-B1, now in the cleanroom at Europe's Spaceport (Arianespace). For VA271, the Ariane 6 core and its

four P160C boosters were moved to the launch pad last week (Arianespace).



Launch sites with orbital launches dated in Launch Library 2 through late November; numbers count launches. Filled: dated launch (Confirmed or Tentative). Hollow: Preliminary. Jiuquan and Vandenberg also flew since Monday.

MAP: THE LAUNCH REPORT · PAD COORDINATES FROM LAUNCH LIBRARY 2 · NATURAL EARTH

III · TRAJECTORIES & HAZARD AREAS

Vulcan heads up the East Coast; Starship's day in orbit

The FAA's operations plan lists the times U.S. airspace is set aside; it gives no hazard-area coordinates.

Airspace windows

The FAA Air Traffic Control System Command Center's plan for Thursday lists these launch windows, converted from UTC (FAA advisory):

FLIGHT	PRIMARY WINDOW	BACKUP
"KV-01" (Vulcan Leo 1)	**Fri Nov 13, 12:54–2:00 p.m.**	**Sat Nov 14, 12:32–1:38 p.m.**
Starship "FLT-15"	Fri Nov 13, 5:45–7:40 p.m.	Sat Nov 14, 5:45–7:40 p.m.

Vulcan's path

"The rocket will launch northeastward from Cape Canaveral, heading up the U.S. East Coast to reach a low Earth orbit," ULA posted, citing a 90 percent chance of acceptable weather from the Space Force's Space Launch Delta 45. A launch readiness review with Amazon, the Space Force and the FAA declared the rocket "GO for liftoff Friday at 12:54 p.m. EST" (ULA), and Thursday is "roll day" (ULA).

Flight 15's timeline

Booster 22 is due back for a catch about 6.5 minutes after liftoff. Ship 42 is to deploy 26 Starlink V3 satellites, circle Earth 14 times and return for its own catch about 22 hours later, Space.com reported. A launch at the opening of the window would bring the ship down at Starbase at 4:17 p.m. EST Saturday (Space.com). Three of the Starlink satellites carry cameras to image the ship's heat shield in orbit, as on Flight 14 (NASASpaceflight).

Spectrum's Norwegian window

An aviation notice from Norway's air navigation provider Avinor opens a launch window for Spectrum's third flight from November 26 through February 17, with attempts possible between 3:00 and 4:30 p.m. EST (20:00 to 21:30 UTC) each day, European Spaceflight reported. Isar Aerospace has not announced a launch date.

ESCAPADE slingshots toward Mars; a cargo Dragon packs for home

Two powered Earth flybys sent NASA's twin probes on their way, the station crew loads Dragon for next week's departure, and seven catalogued objects left orbit.

ESCAPADE's powered flybys

The second ESCAPADE spacecraft, Gold, flew past Earth at about 6 a.m. EST Monday, coming within about 1,050 kilometers, a little more than 48 hours after its twin Blue passed at about 1,600 kilometers; both fired their engines during the flybys, SpaceNews reported. "This is the first time NASA has used a powered Earth flyby to propel a spacecraft toward Mars, and we've now executed this maneuver twice, flawlessly, in just two days," said Asal Naseri, acting associate flight director in NASA's heliophysics division. Rocket Lab, which built both spacecraft, wrote that "Blue and Gold are on their way to Mars!" The pair are due at Mars on Sept. 7 and Sept. 9, 2027 (NASA).

Station: Dragon departure next week



A SpaceX cargo Dragon backs away from the International Space Station after undocking from the Harmony module's forward port on June 16, 2026, above a cloudy North Pacific.

PHOTO: NASA · PUBLIC DOMAIN

The Expedition 75 crew spent Tuesday preserving microbiology samples that "are returned to Earth aboard a SpaceX Dragon cargo spacecraft next week for analysis," and moving cargo in and out of Dragon ahead of its departure, NASA said. NASA's Anil Menon and the Canadian Space Agency's Joshua Kutryk processed bacteria samples in the Kibo laboratory, while station commander Pyotr Dubrov and Anna Kikina serviced three Russian Orlan spacesuits in the Poisk module. The crew is also clearing space for supplies arriving on Northrop Grumman's Cygnus XL in a few weeks. On the schedule, Cygnus NG-25 is set to launch at 10:49 a.m. November 29, after Progress MS-36 at 12:03 a.m. November 24 and China's Shenzhou 24 to Tiangong NET November 24 (Launch Library 2).

Confirmed reentries

CelesTrak's satellite catalog lists these objects as decayed since Monday's issue. Dates are UTC.

OBJECT (CATALOG NO.)	LAUNCHED	DECAYED	LAST ORBIT, KM
Long March 4 debris (20893)	**Sep 3, 1990**	**Nov 9**	**302 × 313**
Starlink-5701 (55487)	Feb 2, 2023	Nov 9	145 × 152
Starlink-1274 (45387)	Mar 18, 2020	Nov 10	152 × 156
Starlink-1400 (45700)	Jun 4, 2020	Nov 10	162 × 166
Starlink-30221 (57418)	Jul 20, 2023	Nov 10	210 × 216
Starlink-32264 (61642)	Oct 23, 2024	Nov 10	184 × 222
Starlink-1284 (45390)	Mar 18, 2020	Nov 11	145 × 149

Six of the seven are Starlink satellites, the oldest from March 2020. The seventh is a fragment from a Chinese Long March 4 rocket launched in September 1990, which spent 36 years in an orbit inclined 98 degrees before it decayed (CelesTrak).

Booster 22 rolls out; Neutron's price goes up

SpaceX readies the first double catch, the FAA moves Starbase hearings online, Rocket Lab raises prices on strong demand, and NASA explains its plan for SLS.

SpaceX: Booster 22 on the pad

"Super Heavy moved to the launch pad at Starbase. Booster 22 is the first Super Heavy V3 booster planned to return to the launch site for catch and was used for final catch testing upon arrival, including a suspended squeeze test at catch height," SpaceX posted at 1:21 p.m. EST Tuesday; Elon Musk (@elonmusk) shared it Wednesday. Booster 22 and Ship 42 are both on their first flights, and Flight 15 will be Starship's 15th full-stack launch and the fourth for the V3 version, with the first booster catch attempt since Flight 8 (NASASpaceflight). SpaceX has caught Super Heavy three times but has never tried to catch a ship (Space.com).

ON THE WIRE

Spectrum's long window. Isar Aerospace's third Spectrum can fly any day from November 26 through February 17 under Avinor's aviation notice (European Spaceflight).



The launch tower's arms catch a Super Heavy booster at Starbase, Texas, during Starship's fifth flight test on Oct. 13, 2024. No ship has been caught yet.

PHOTO: STEVE JURVETSON, "STARSHIP BOOSTER LANDING ON MECHZILLA," WIKIMEDIA COMMONS · CC BY 2.0 (CROPPED)

FAA: Starbase hearings go online

The FAA has moved all public scoping meetings on SpaceX's plan for up to 456 Starship launches a year at Starbase online, including those set for November 17 and 19, and extended the comment period to December 7 (FAA). Five virtual meetings run November 17 to 19, at 2:00 or 6:30 p.m. EST. Tuesday's Federal Register notice (docket FAA-2026-12808) says later Starship license modifications or renewals fall within the scope of the programmatic environmental impact statement.

ULA: Vulcan's comeback flight



Vulcan lifts off from Space Launch Complex 41 at Cape Canaveral Space Force Station on its second certification flight on Oct. 4, 2024.

U.S. SPACE FORCE PHOTO BY AIRMAN 1ST CLASS COLLIN WESSON · PUBLIC DOMAIN

Friday's flight debuts the VC6L, with six solid rocket boosters and the long payload fairing, and a low-Earth-orbit version of the Centaur V upper stage that is 1.94 meters smaller than the High Energy version flown on every earlier Vulcan, NASASpaceflight reported. It is Vulcan's fifth mission, a certification flight of the configuration for the U.S. Space Force, and the first of 38 Vulcan launches scheduled for Amazon Leo. A dedicated processing lane with purpose-built infrastructure for Amazon's constellation "is ready to support its first launch at Cape Canaveral," ULA posted Thursday.

Rocket Lab: Neutron at $70 million

Rocket Lab raised the price of a dedicated Neutron launch from $55 million to $70 million, citing strong demand and a tight launch market, and said it booked eight new Neutron missions in the third quarter worth more than $515 million, doubling the rocket's backlog, Payload reported. Neutron's first launch has slipped to 2027, and the company also expects to raise Electron and HASTE prices. "Electron is all dressed up for launch 100," the company posted Tuesday.

QUICK HIT

United Kingdom. The government now holds more than 25 percent, but not more than 50 percent, of SaxaVord Spaceport's operating company after a £60 million public-private investment, according to Companies House filings (European Spaceflight).

NASA: the SLS road map

NASA Administrator Jared Isaacman said industry responses on future commercial lunar transportation point to the 2030s, so the agency will standardize the Space Launch System (SLS), move to a stretched Centaur V upper stage and stop work on Mobile Launcher 2, NASASpaceflight reported. "There are a lot of ways to shake the $4 billion disposable rocket label that SLS has carried and one of which is launch more frequently," he said. Both the last Interim Cryogenic Propulsion Stage and a stretched Centaur V remain options for Artemis IV.

Europe: MaiaSpace's contract

The European Space Agency (ESA) has not announced signing MaiaSpace's European Launcher Challenge contract because the company alone chose support through launch service contracts, worth €183.4 million, rather than development co-funding; ESA signed €543.6 million in contracts with Isar Aerospace, Rocket Factory Augsburg and PLD Space in August (European Spaceflight).

Still open

Comments on the Federal Communications Commission's space-licensing overhaul (SB Docket No. 25-306) are due November 25 (Federal Register), and comments on the FAA's three latest proposals to ease Part 450, its launch and reentry licensing rule, are due December 9 (Federal Register).

ON THE WIRE

Friday's forecast. The Space Force's Space Launch Delta 45 weather squadron gives a 90 percent chance of acceptable conditions for Vulcan Leo 1 (ULA).

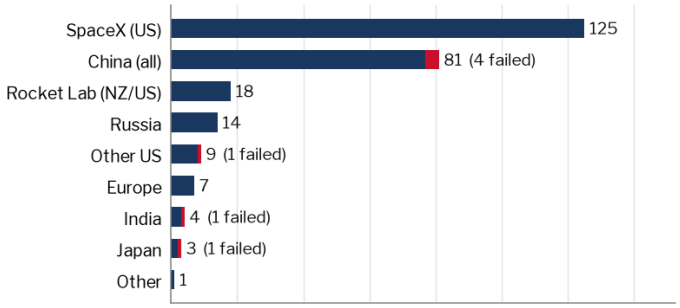
262 orbital launch attempts so far in 2026

Two successes since Monday, added from Launch Library 2 to GCAT's count through Sunday's Long March 12.

262	255	125	81	18
orbital launch attempts worldwide through Nov 11	reached orbit	by SpaceX, all successful	by China, four failures	by Rocket Lab, all successful

The count

Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) records 260 orbital launch attempts in 2026 through Sunday; with Wednesday's Long March 2D and Falcon 9 from Launch Library 2, the year stands at 262, 255 of them successful. SpaceX leads with 125, followed by China with 81, Rocket Lab with 18 and Russia with 14; other U.S. providers have flown nine, Europe seven, India four, Japan three and South Korea one.



Orbital launch attempts in 2026 by provider or country. Navy: success. Red: failure or partial failure. "Other" is South Korea.

CHART: THE LAUNCH REPORT · GCAT THROUGH NOV 8, PLUS LAUNCH LIBRARY 2

The failures

According to GCAT, seven launches have failed this year: India's PSLV on January 12; China's Long March 3B on January 16, Ceres-2 on January 17, Tianlong-3 on April 3 and Long March 7A on August 10; Japan's Kairos on March 5; and Blue Origin's New Glenn on April 19, recorded as a partial failure.

The busiest sites

Orbital launch attempts in 2026 by site (GCAT through Sunday, plus Wednesday's two from Launch Library 2):

SITE	ATTEMPTS
Vandenberg, Calif.	**71**
Cape Canaveral, Fla.	56
Jiuquan, China	31
Mahia, New Zealand	17
Hainan commercial site, Wenchang, China	16
Taiyuan, China	13
Plesetsk, Russia	9

The week ahead

Six orbital launches have specific dates between Thursday and next Wednesday (Launch Library 2; FAA): a Kuaizhou 11 this afternoon, Vulcan Leo 1 and Starship Flight 15 on Friday, Electron's 100th mission early Saturday, USSF-317 late Saturday and USSF-266 on Wednesday.

Takeaways

1. Both launches since Monday succeeded, bringing 2026 to 262 attempts and 255 successes.
2. Vulcan returns Friday at 12:54 p.m. with 40 Amazon Leo satellites, its first flight in more than nine months.
3. Starship Flight 15 holds FAA airspace windows from 5:45 p.m. Friday for the first try at catching both stages.
4. NASA's ESCAPADE probes completed two powered Earth flybys and are headed for Mars.
5. Starbase scoping comments now run to December 7, with all meetings online.

KEY DATES AHEAD (EST)

Thu Nov 12	Kuaizhou 11, 2:50 p.m.
Fri Nov 13	Vulcan Leo 1, 12:54 p.m.; Starship Flight 15, 5:45 p.m.
Sat Nov 14	Electron's 100th mission, 12:30 a.m.; USSF-317, 10:55 p.m.
Nov 17–19	Virtual Starbase scoping meetings
Wed Nov 18	USSF-266, 11:49 a.m.
Tue Nov 24	Progress MS-36, 12:03 a.m.
Mon Dec 7	Starbase scoping comments due

SOURCES

Schedules: Launch Library 2 (The Space Devs), NASASpaceflight, FAA operations plan advisory. Agencies: NASA, FAA, FCC, Federal Register. Reporting: Space.com, SpaceNews, NASASpaceflight, Payload, European Spaceflight, Korea JoongAng Daily, SBS. Tracking: CelesTrak, GCAT (Jonathan McDowell). X posts: @SpaceX, @elonmusk, @ulalaunch, @RocketLab, @Arianespace, @planet4589. Photos: NASA (public domain); Steve Jurvetson (CC BY 2.0). Maps: Natural Earth (public domain).

The Launch Report is compiled from official agency releases, regulatory filings, orbital tracking data, public posts on X and published reporting. Quotes are reproduced verbatim unless marked as translated or paraphrased. Launch times change often; check with the launch provider before making plans.